

Experiment No. 7

Lab Practical: Application Layer Packet Analysis using Packet Tracer Tool

Aim:

To analyze Application Layer packets (DNS & HTTP) using a packet tracer tool (Wireshark) and understand their structure and working.

Tools Required:

- Wireshark (Packet Analyzer)
- Internet Connection or Local DNS/HTTP Server
- Browser or Command Line Tools (nslookup, ping)

Procedure:

Part A: DNS Packet Analysis

1. **Start Wireshark** and begin packet capture on the active network interface (WiFi/Ethernet).
2. Open **Command Prompt** or **Terminal** and type:
3. `nslookup www.google.com`
4. In Wireshark, stop the capture after response.
5. Apply filter:
6. `dns`
7. Select the DNS query and observe the following:
 - **Transaction ID**
 - **Flags:** Query/Response
 - **Query Name:** e.g., `www.google.com`
 - **Query Type:** A (IPv4)
8. Click on the DNS **Response** packet and observe:
 - **Resolved IP Address**
 - **TTL** (Time to Live)
 - **Answer Section**

Part B: HTTP Packet Analysis

1. In the browser, type:
2. `http://example.com`
3. In Wireshark, apply filter:
4. `http`
5. Observe the **HTTP GET** request:
 - **Method:** GET

- **Host:** example.com
- **User-Agent**
- 6. Observe the **HTTP Response:**
 - **Status Code:** 200 OK
 - **Content-Type:** text/html
 - **Content-Length**

Observations:

Field	DNS Query	DNS Response	HTTP Request	HTTP Response
Protocol	UDP	UDP	TCP	TCP
Port	53	53	80	80
Method/Type	Standard Query (A Record)	Response	GET /	200 OK
IP Address Queried	www.google.com	e.g., 142.250.77.68	example.com	IP of example.com
TTL	-	300 seconds (example)	-	-
Content-Type	-	-	-	text/html

Result:

Successfully analyzed DNS and HTTP packets using Wireshark. The structure and fields of Application Layer packets were observed and interpreted.

Conclusion:

- DNS resolves domain names to IP addresses via **UDP port 53**.
- HTTP works over **TCP port 80**, where the client sends a GET request and receives a response with status and content.
- Wireshark effectively captures and visualizes **Application Layer** communication for analysis and troubleshooting.

Output / Screenshots

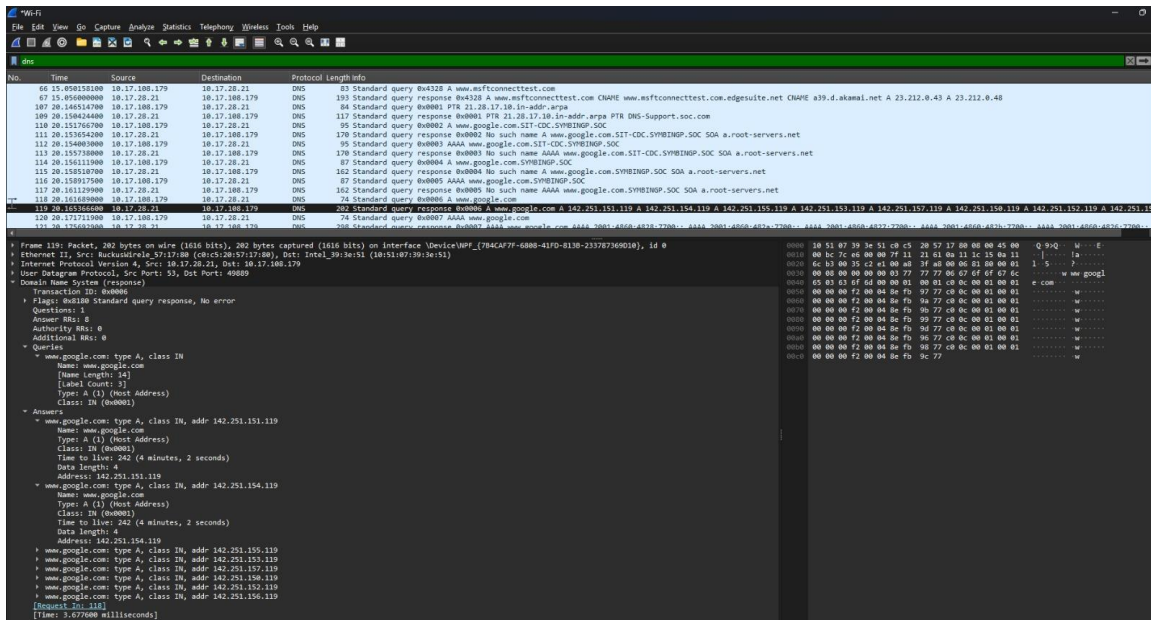


Fig 1

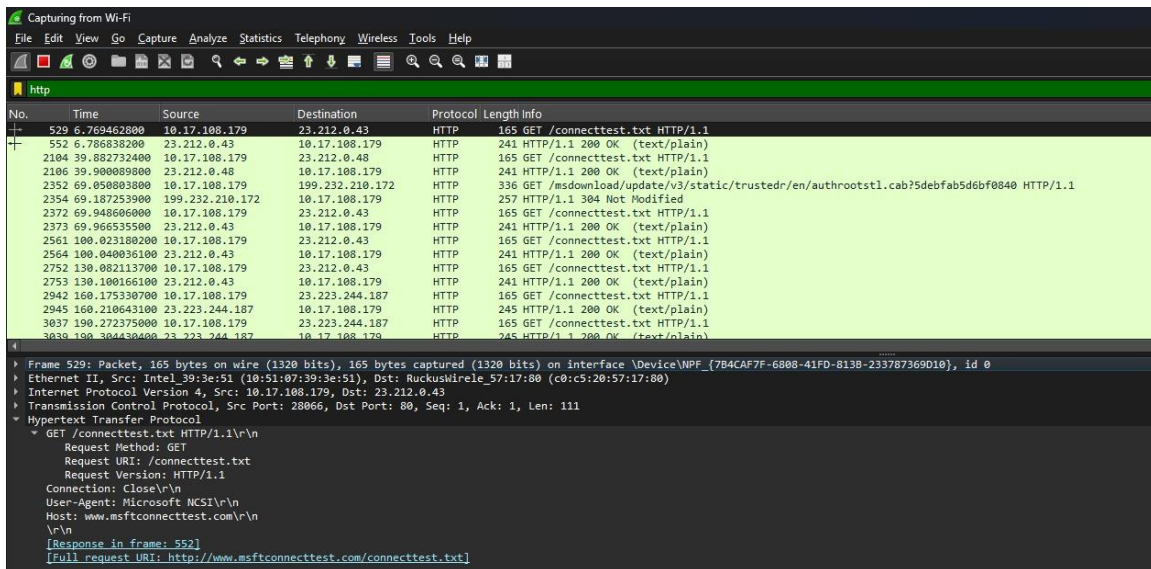


Fig 2

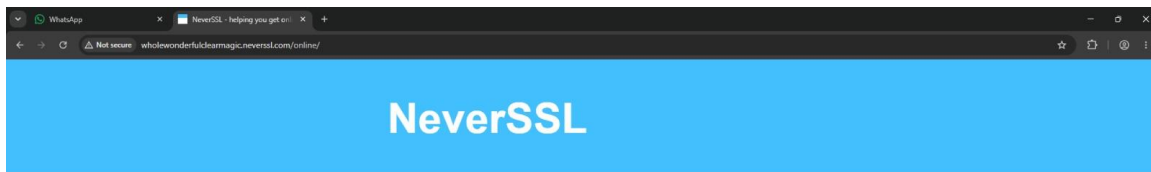


Fig 3